

Project Summary x Schematic x universaltobasic_tb.v x universaltobasic.v x

C:/Users/P.SATHWIKA/Implement/Implement.srscs/sources_1/new/universaltobasic.v



14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 //////////////////////////////////////
21
22
23 module universaltobasic(
24 input a,
25 input b,
26 output y_not,
27 output y_and,
28 output y_or
29);
30
31 wire w_nota, w_notb, w_and_internal;
32
33
34 nand u_not (y_nota, a, a);
35
36
37 nand u_and_1 (w_and_internal, a, b);
38 nand u_and_2 (y_and, w_and_internal, w_and_internal);
39
40
41 nand u_or_not1 (w_nota, a, a);
42 nand u_or_not2 (w_notb, b, b);
43 nand u_or_comb (y_or, w_nota, w_notb);
44
45
46 assign y_not = y_nota;
47
48 endmodule
49

<

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16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module universaltobasic_tb;
24     reg a,b;
25
26     wire y_not;
27     wire y_and;
28     wire y_or;
29
30     // Instantiate the single unified module
31     gates_using_nand uut (
32         .a(a),
33         .b(b),
34         .y_not(y_not),
35         .y_and(y_and),
36         .y_or(y_or)
37     );
38
39     initial begin
40         $display("Time\t A \t B \t NOT(A)\t AND\t OR");
41         $monitor("%0t\t %b \t %b \t %b \t %b \t %b \t %b", $time, a, b, y_not, y_and, y_or);
42
43         a = 0; b = 0; #10;
44         a = 0; b = 1; #10;
45         a = 1; b = 0; #10;
46         a = 1; b = 1; #10;
47
48         $finish;
49     end
50 endmodule
51

```

```
# send_msg_id Add_Wave-1 WARNING "No top level signals found. Simulator will start without a wave window. If you want to open a wave window go to 'File->New Waveform ('
# }
# }
# run 1000ns
Time      A      B      NOT(A)  AND      OR
0         0      0      1         0         0
10000     0      1      1         0         1
20000     1      0      0         0         1
30000     1      1      0         1         1

$finish called at time : 40 ns : File "C:/Users/P.SATHWIKA/implment/implment.srcs/sim_1/new/universaltobasic_tb.v" Line 48
INFO: [USF-XSIM-96] XSim completed. Design snapshot 'universaltobasic_tb_behav' loaded.
INFO: [USF-XSIM-97] XSim simulation ran for 1000ns
launch_simulation: Time (s): cpu = 00:00:06 ; elapsed = 00:00:14 . Memory (MB): peak = 1263.211 ; gain = 0.000
```

